



# Gunateja Pothula

## Data Analyst/ Data Scientist

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🔗 [gunateja020.github.io/Portfolio](https://gunateja020.github.io/Portfolio)

Recent Master's graduate in Data Science with expertise in machine learning and time-series forecasting. Developed XGBoost and LSTM models on 140K+ aviation records for flight delay prediction, achieving projected cost savings of \$140K+. Proficient in Python, SQL, database design, and end-to-end model development. Seeking Data Scientist or Machine Learning Engineer roles.

## Education

**Masters in Data Science, Arden University** 11/2024 – 11/2025  
Berlin, Germany

**Thesis: Flight Delay Prediction & Optimization Using ML and Time-Series Analysis**

- Developed and evaluated **XGBoost and LSTM time-series models** on 140K+ flight records to predict departure delays, applying concepts from **Machine Learning and Artificial Intelligence and Neural Networks**.
- Performed extensive **feature engineering, hyperparameter tuning, and cross-validation** to optimize predictive performance and model reliability.
- Processed and managed large aviation datasets using **Python (Pandas, NumPy)** and scalable data-processing approaches aligned with **Big Data and Cloud Computing** principles.
- Applied statistical analysis and evaluation metrics (**RMSE, MAE, Accuracy**) using techniques from **Mathematics for Data Science** to compare model effectiveness.
- Conducted **exploratory data analysis (EDA)** to identify patterns related to weather conditions, airport congestion, and operational factors influencing delays.
- Created analytical dashboards and visualizations using **Tableau and Python libraries (Matplotlib, Seaborn)**, applying concepts from **Data Visualisation**.
- Built end-to-end data processing and modeling workflows using **Python**, demonstrating skills developed in **Programming for Data Analytics**, and produced forecasting insights supporting data-driven scheduling decisions with projected cost savings exceeding **\$140K**.

**Bachelors in Communications, Visvesvaraya Technological University** 06/2019 – 08/2023  
India

**Academic Project:**

- Designed and implemented a real-time face recognition system using computer vision techniques, achieving 95%+ accuracy.
- Automated secure access for 100+ users, eliminating manual identity verification.

## Professional Experience

**Programmer Analyst Trainee, Cognizant Technology Solutions** 02/2024 – 07/2024  
India

- Designed and implemented MongoDB database schemas for healthcare patient management system handling thousands of records
- Developed and optimized database queries for efficient data retrieval, filtering, and aggregation across patient datasets
- Implemented data validation logic and quality checks ensuring 99.9% data integrity across the application
- Collaborated with cross-functional teams to translate business requirements into scalable database solutions

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| <b>Machine Learning Engineer, <i>Freelance</i></b> <ul style="list-style-type: none"> <li>Developed and deployed 5+ end-to-end machine learning solutions for sports analytics</li> <li>Engineered deep learning models (Random Forest, XGBoost, CNN) for Formula 1 race prediction achieving F1-score <math>\geq 0.8</math> and RMSE <math>&lt; 10</math></li> <li>Designed interactive dashboards using Streamlit, Plotly Dash, and Tableau for real-time data visualization and business intelligence</li> <li>Implemented automated data pipelines processing 100K+ records daily with 99.5% uptime</li> <li>Applied unsupervised learning (K-Means, GMM) for customer segmentation and pattern recognition</li> </ul> | 05/2023 – 01/2024<br>India |
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| <b>Internship Trainee, <i>Pentagon Space</i></b> <ul style="list-style-type: none"> <li>Developed a Java-based full-stack banking application managing 100+ customer accounts and transactions</li> <li>Designed SQL schemas and implemented validation logic to ensure data accuracy</li> </ul> | 02/2023 – 03/2023<br>India |
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| <b>Data Science Research Assistant, <i>Visvesvaraya Technological University</i></b> <ul style="list-style-type: none"> <li>Conducted research on deep learning applications for predictive analytics and time series forecasting</li> <li>Developed Python-based data analysis scripts for processing academic research datasets with 10K+ samples</li> <li>Created data visualizations and statistical reports</li> <li>Collaborated with faculty on machine learning projects involving regression, classification, and clustering algorithms</li> <li>Presented research findings at departmental seminars and contributed to technical documentation</li> </ul> | 06/2022 – 01/2023<br>India |
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## Skills

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| <b>Programming &amp; ML</b> — C, C++, Python (pandas, NumPy, scikit-learn, TensorFlow, Keras), Java                                                                          |
| <b>Machine Learning &amp; Data Analysis</b> — XGBoost, LSTM, Time-Series Forecasting, Feature Engineering, Regression, Classification, Model Evaluation, Clustering, PyTorch |
| <b>Data Visualization</b> — Tableau, Matplotlib, Seaborn, Plotly                                                                                                             |
| <b>Databases &amp; Tools</b> — MongoDB, MySQL, PostgreSQL, Git, Docker                                                                                                       |
| <b>Web &amp; APIs</b> — HTML, CSS, JavaScript, MERN Stack (MongoDB, Express, React, Node.js), REST APIs                                                                      |
| <b>AI Tools</b> — Large language models , Prompt Engineering , Context Engineering, LLM APIs , RAG                                                                           |

## Languages

**English** — Fluent
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 **Hindi** — Fluent
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 **German** — B1